AGRICULTURE INFORMATION PORTAL (SRS)
(WSD PROJECT)

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1.0 Introduction

1.1 Purpose

The primary objective of this document is to clearly define the software requirements of Agriculture Information Portal. In simple terms, it is a web-based portal designed to provide farmers with easy and immediate access to vital information related to agriculture. In this document, you will be able to find functional and non-functional requirements, target audience, and the technology stack that we will use for its development.

1.2 Scope

This website can be thought of as the one-stop destination for any person who wants to know about farming, whether it is an ordinary farmer or an agriculture student. One can simply visit the website and go through the information about various crops grown in India, along with tips on cultivation, weather information, fertilizers needed, and information about various schemes offered by the government. For the time being, the website is more of a static front-end without any heavy server-side processing or user login facility.

1.3 Stack Description

- **Implementation as of now:**
- HTML5 for the core structure of our webpage, tables, lists and forms.
- Inline CSS3 for managing all the styles, color schemes and alignments.
- Basic HTML forms for the Farmer Registration section.
- **What needs to be included in the final build?**
- Backend scripts (such as PHP, Node.js) for processing the form.
- Relational database management system (MySQL is recommended) for storing the registration information of the users.
- Client-side JavaScript for validating forms and fetching real time data like weather data via APIs.

1.4 Definitions, Acronyms and Abbreviations

Term	Description
SRS	Software Requirements Specification
Kharif	Cropping season, June-November (monsoon)
Rabi	Cropping season, November-April (winter)
NPK	Nitrogen, Phosphorus, Potassium fertilizer
DAP	Di-Ammonium Phosphate fertilizer
PM-KISAN	Pradhan Mantri Kisan Samman Nidhi scheme
e-NAM	National Agriculture Market, online trading platform
UI / DFD	User Interface / Data Flow Diagram

2.0 Overall Description

2.1 Requirement of this Web Application

One major challenge that the farmers of India face is that there is crucial information related to agriculture which can be found scattered in numerous websites of the government as well as private sector. This becomes extremely difficult to search through. Our web app solves this exact issue by bringing all the relevant information in one simple and user-friendly website which even supports mobile devices.

2.2 Module Description

The application has been divided into the following sections :

• Home / Navigation: This consists of the main banner and an organized navigation bar that allows one to navigate through the portal.

• About Us: This is a short description about the purpose of the portal

• Crops: This describes the major crops such as Rice, Wheat, Sugarcane and Cotton

• Crop Information: It is an information table regarding the season of production, type of soil required, and market price.

• Weather Information: This provides the temperature, humidity, and amount of rainfall in certain agricultural regions.

• Fertilizer Recommendation: This shows the relation between crops and fertilizers and why certain fertilizers must be used for particular crops.

• Government Schemes: The schemes provided by the government are explained here, including PM-KISAN and Soil Health Card scheme.

• Farmer Registration: This is an input form that helps farmers subscribe for alerts.

• Contact Us: Contact us page.

2.3 Sub-Module Description

Farmer Registration Module includes the following components:

• Personal Information: Input fields for Name, Email, Mobile Number, and State.

• Crops Selection: Dropdown menu to select their main crop and radio buttons to select their gender.

• Services Selection: Check boxes to select the services they prefer (such as Weather, Market Rates, and Schemes).

• Form Action: Default Submit and Clear buttons for handling form data.

3.0 Functional Requirements

3.1 Functional Requirements List

• The navigation links need to be functional, allowing users to move directly to different sections.

- The application should provide a list of the crops supported by the website and descriptions of them.

- The application should present a properly formatted table with information about planting seasons, soil type required, and prices of different crops currently available.

- The application should display the weather in a particular area according to the location selected.

- The application should provide an overview of different fertilizers suggested for different crops.

- The website should provide the list of the government schemes currently active and describe them.

- A working registration form should be provided, collecting data about the user and services they need.

- Users should be able to clear and submit the registration form without page malfunction.

3.2Software and Hardware Requirements

Client Side (End User):

- Operating System: Any recent Operating Systems like Windows, Mac OS, Android, or iOS.
- Browser: Recent versions of Chrome, Firefox, Safari, or Edge.
- Hardware: Essentially, any device (smartphone/computer) capable of web browsing.
- Connectivity: Internet Connection should be functional for accessing external resources and API calls.

Server Side (Hosted Environment):

- Operating System: Linux (Ubuntu) or Windows Server.
- Web Server: Apache/Nginx.
- Runtime/Database: PHP 8.x (or Node.js) & MySQL 8.x.
- Required Hardware: 1 vCPU, 2 GB RAM, and at least 5 GB SSD Disk space for storage of files.

4.0 Table Description and Data Flow Diagram

4.1 Table Description (Database Schema)

Here is the database schema showing each module, its sub-modules, purpose, and the user type that can access it:

Module	Sub-Module	Description	User Type
Home / Navigation	—	Main banner and navigation bar for browsing key sections of the portal.	Visitor
About Us	—	Overview of the portal and how it helps farmers access agricultural info.	Visitor
Crops	—	Detailed list of major Indian crops (Rice, Wheat, Sugarcane, Cotton) with brief notes.	Visitor
Crop Information	—	Data table of growing seasons, soil types, and average market prices.	Visitor
Weather Information	—	Current temperature, humidity, and rainfall for selected farming areas.	Visitor
Fertilizer Recommendation	—	Maps crop to recommended fertilizers and explains their application purpose.	Visitor
Government Schemes	—	Breakdown of central schemes such as PM-KISAN and the Soil Health Card.	Visitor
Farmer Registration	Personal Details	Inputs for name, email, mobile number, and state.	Visitor
Farmer Registration	Crop Selection	Dropdown list for primary crop and radio buttons for gender.	Visitor
Farmer Registration	Service Preference	Checkboxes to select updates (Weather, Market Prices, Schemes).	Visitor
Farmer Registration	Form Action	Standard Submit and Clear buttons to handle the form data.	Visitor
Contact Us	—	General contact details for support or questions.	Visitor
Admin Panel	Content Management	Manage crop, weather, fertilizer & scheme content and pricing.	Admin

4.2 Primary Scenarios of Usage

• A farmer finds out the fertilizers that will be required for a certain crop immediately before the start of planting period.

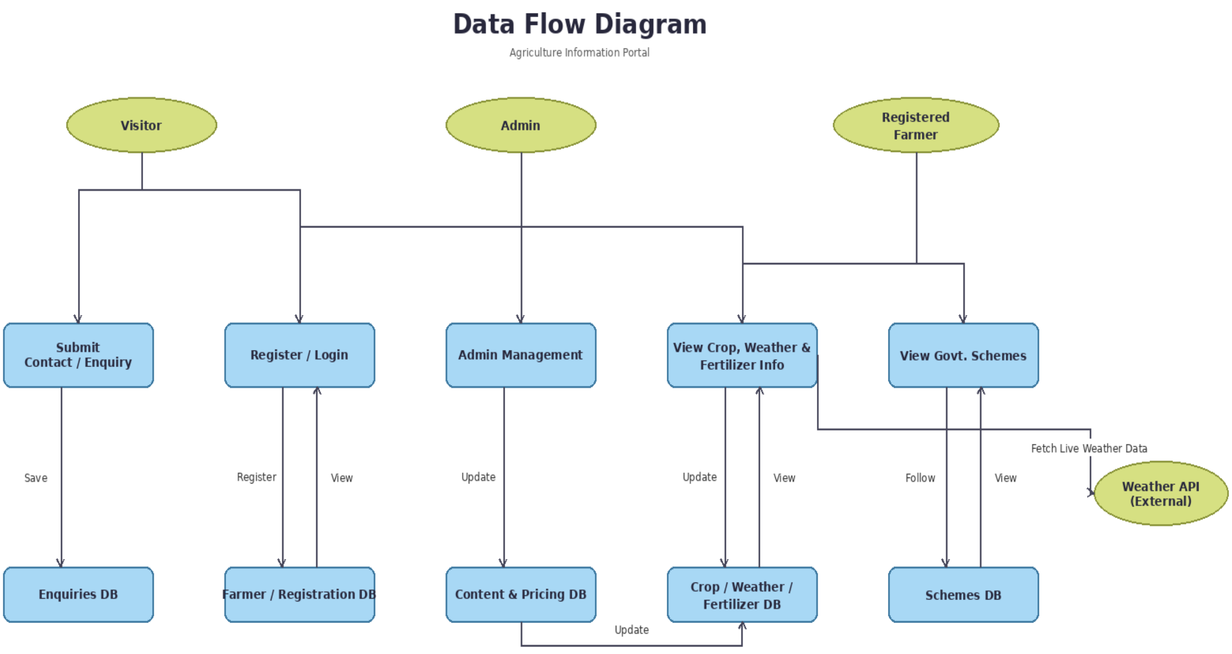
• The user seeks weather forecasts of the area prior to irrigating their lands.

• A farmer signs into the portal so as to receive daily notifications regarding the prices of crops on their mobile phones.

• Student or visitor gets information on the recently introduced government schemes.

4.3 Data Flow Diagram (DFD)

Description of Data Flows:



5.0 References

1. **W3C HTML & CSS Standards**
   Official guidelines for web structuring and styling.
   https://www.w3.org/

2. **Department of Agriculture & Farmers Welfare, India** Used for referencing valid government schemes and crop data.
   https://agricoop.nic.in/

3. **PHP and MySQL Documentation**
   Backend scripting and database management guidelines.
   https://www.php.net/manual/en/

4. **Indian Meteorological Department (IMD)**
   Reference for formatting and gathering agricultural weather data.
   https://mausam.imd.gov.in/